Physical Chemistry (I)

Lecture 05

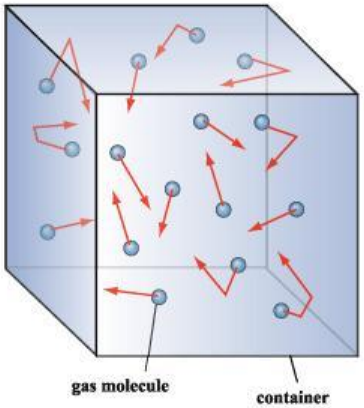
Real Gas

First Law of Thermodynamics

In the Last Class

- Kinetic molecular theory
  - Pressure: perfect gas law
  - Internal energy
- Equipartition Theorem
- Real gas
  - Compression factor
  - Virial expansion

Pressure



momentum change ⟶ force ⟶ pressure

$$pV = \frac{1}{3} nM \langle v^2 \rangle$$

$$pV = nRT$$

(Image: http://web2.uwindsor.ca/courses/physics/high_schools/2005/Brownian_motion/kinetictheory.jpg)

Internal Energy, *U*

"The total energy of a system"

For a perfect gas,

$$U = \sum_{i=1}^{N} \frac{1}{2} m v_i^2 = \frac{1}{2} M n \langle v^2 \rangle$$

$$U = \frac{3}{2} nRT$$

Last class

Equipartition Theorem

For a sample at thermal equilibrium,

the average value of each quadratic contribution

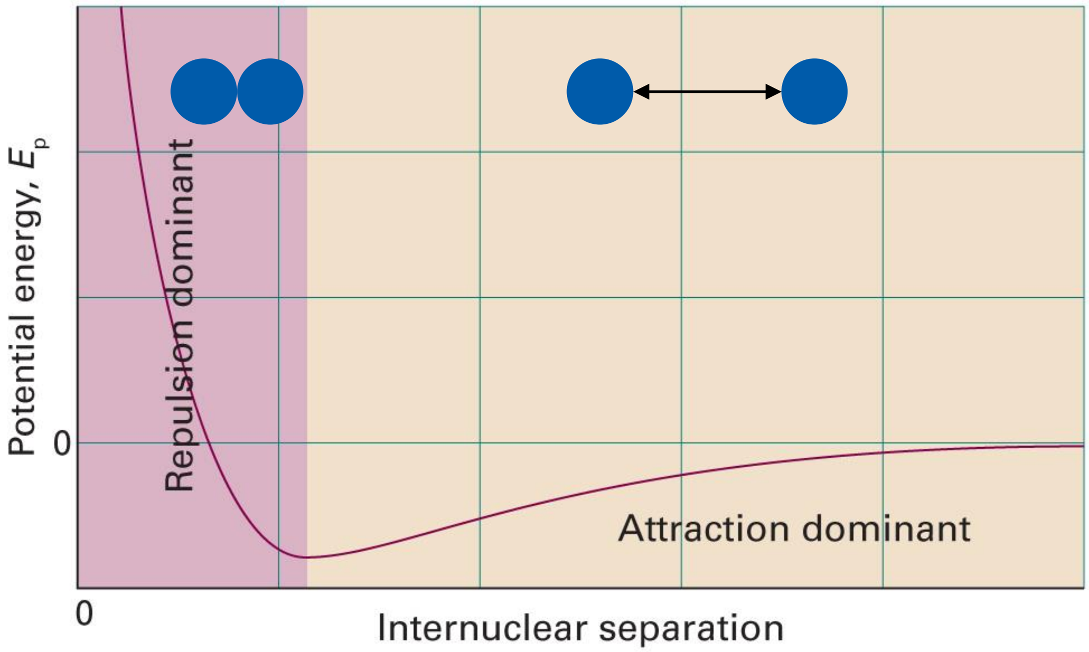
to the internal energy is $(1/2)k_{\mathrm{B}}T$.

(Quadratic contribution: $E \propto p_x^2$ or $E \propto x^2$)

Translation + Rotation + Vibration

Real Gas

Now the molecules interact!



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 1C.1)

Last class

Compression Factor Z

$$Z = V_m/V_m^{o}$$

Molar volume of a gas $V_m = V/n$

Molar volume of a perfect gas V_m^{o}

- $Z > 1$: $V_m > V_m^{o} \rightarrow$ repulsion is dominant
- $Z \approx 1$: $V_m \approx V_m^{o} \rightarrow$ perfect-gas-like behavior
- $Z < 1$: $V_m < V_m^{o} \rightarrow$ attraction is dominant

Last class

Virial Expansion

$$Z(p) = 1 + B'p + C'p^2 + \cdots$$

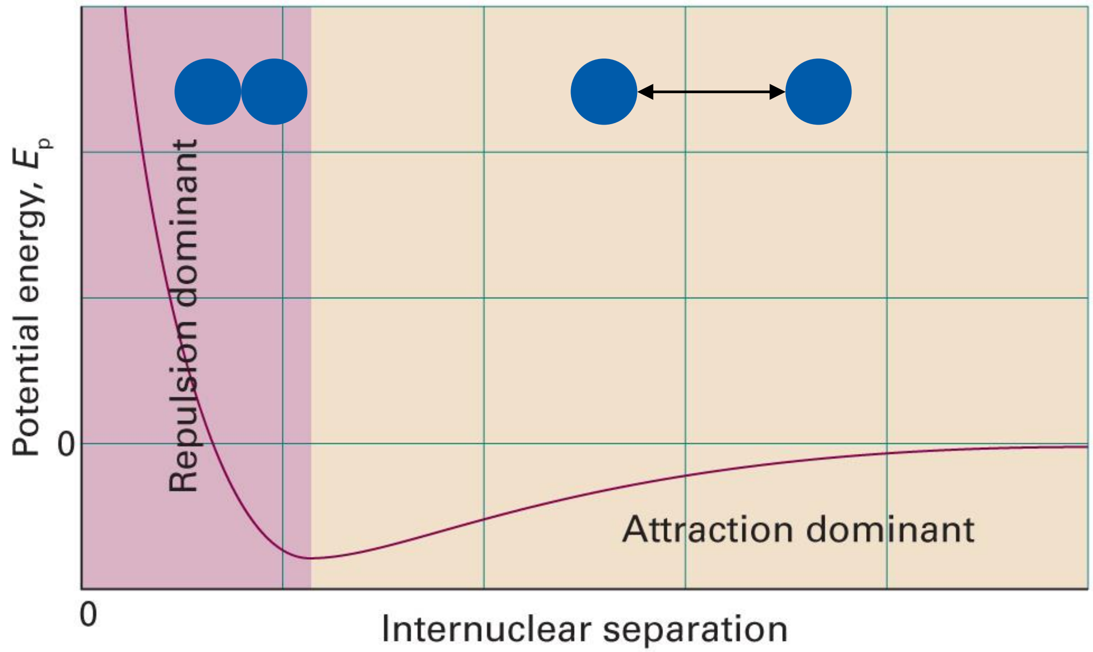
$$Z(V_m) = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \cdots$$

- Second virial coefficient B
  - $B > 0 \rightarrow Z > 1 \rightarrow$ repulsion is dominant
  - $B = 0 \rightarrow Z \approx 1$ (perfect-gas-like) for a larger region
    - Boyle temperature T_{B}: temperature where $B = 0$
  - $B < 0 \rightarrow Z < 1 \rightarrow$ attraction is dominant

Topic 1C

Corrected Perfect Gas Law

Consider the effects of repulsion and attraction



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 1C.1)

van der Waals Equation

- Repulsion: decrease of the total volume

$$V \to V - nb$$

$$p = \frac{nRT}{V - nb}$$

- Attraction: decrease of the pressure

$$p \sim \text{(frequency of collisions)} \times \text{(force of each particle)}$$

decreases as $n/V \uparrow$ decreases as $n/V \uparrow$

$$p = \frac{nRT}{V - nb} - a\frac{n^2}{V^2}$$

van der Waals Coefficients

$$p = \frac{nRT}{V - nb} - a\frac{n^2}{V^2}$$

- a: strength of attractive interactions
- b: strength of repulsive interactions

- Characteristics of each gas
- Independent of the temperature

Experimental Data

	a / (atm L^2 mol^{-2})	b / (10^{-2} L mol^{-1})
Ar	1.337	3.20
CO_2	3.610	4.29
He	0.0341	2.38
Xe	4.137	5.16

Equation Manipulations

$$p = \frac{nRT}{V - nb} - a\frac{n^2}{V^2}$$

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

can be represented by a cubic equation

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van der Waals and Virial

- van der Waals equation

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

- Virial expansion

$$pV_m = RT\left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \cdots\right)$$

Can we connect the two?

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Some Math

Target

$$pV_m = RT\left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \cdots\right)$$

vdW: $p = \dfrac{RT}{V_m - b} - \dfrac{a}{V_m^2}$

$$p = \frac{1}{V_m}\left(\frac{RT}{1 - b/V_m}\right) - \frac{a}{V_m^2}$$

$$pV_m = \frac{RT}{1 - b/V_m} - \frac{a}{V_m}$$

$$pV_m = RT\left(\frac{1}{1 - b/V_m} - \frac{a/RT}{V_m}\right)$$

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Some More Math

Target

$$pV_m = RT\left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \cdots\right)$$

$$pV_m = RT\left(\frac{1}{1 - b/V_m} - \frac{a/RT}{V_m}\right)$$

As $b/V_m \ll 1$, we use

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots \text{(when } x < 1\text{)}$$

$$pV_m = RT\left[\left\{1 + \frac{b}{V_m} + \left(\frac{b}{V_m}\right)^2 + \cdots\right\} - \frac{a/RT}{V_m}\right]$$

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Finally

Target

$$pV_m = RT\left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \cdots\right)$$

$$pV_m = RT\left[\left\{1 + \frac{b}{V_m} + \left(\frac{b}{V_m}\right)^2 + \cdots\right\} - \frac{a/RT}{V_m}\right]$$

$$pV_m = RT\left[1 + \left(b - \frac{a}{RT}\right)\frac{1}{V_m} + \frac{b^2}{V_m^2} + \cdots\right]$$

Hence,

$$B = b - a/RT \text{ and } C = b^2$$

Also, from $B = b - a/RT = 0$, $T_\mathrm{B} = a/(bR)$

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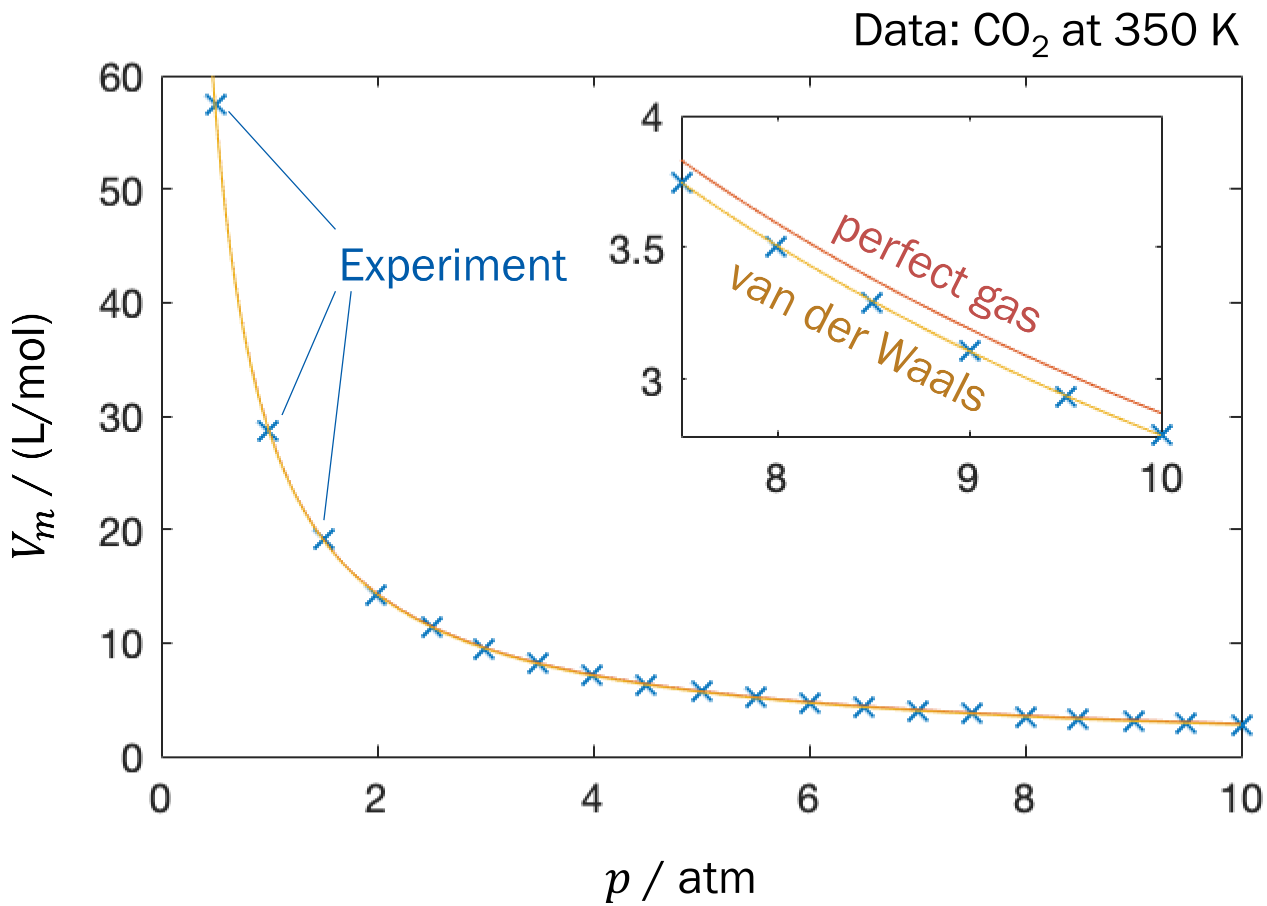
Validity of van der Waals



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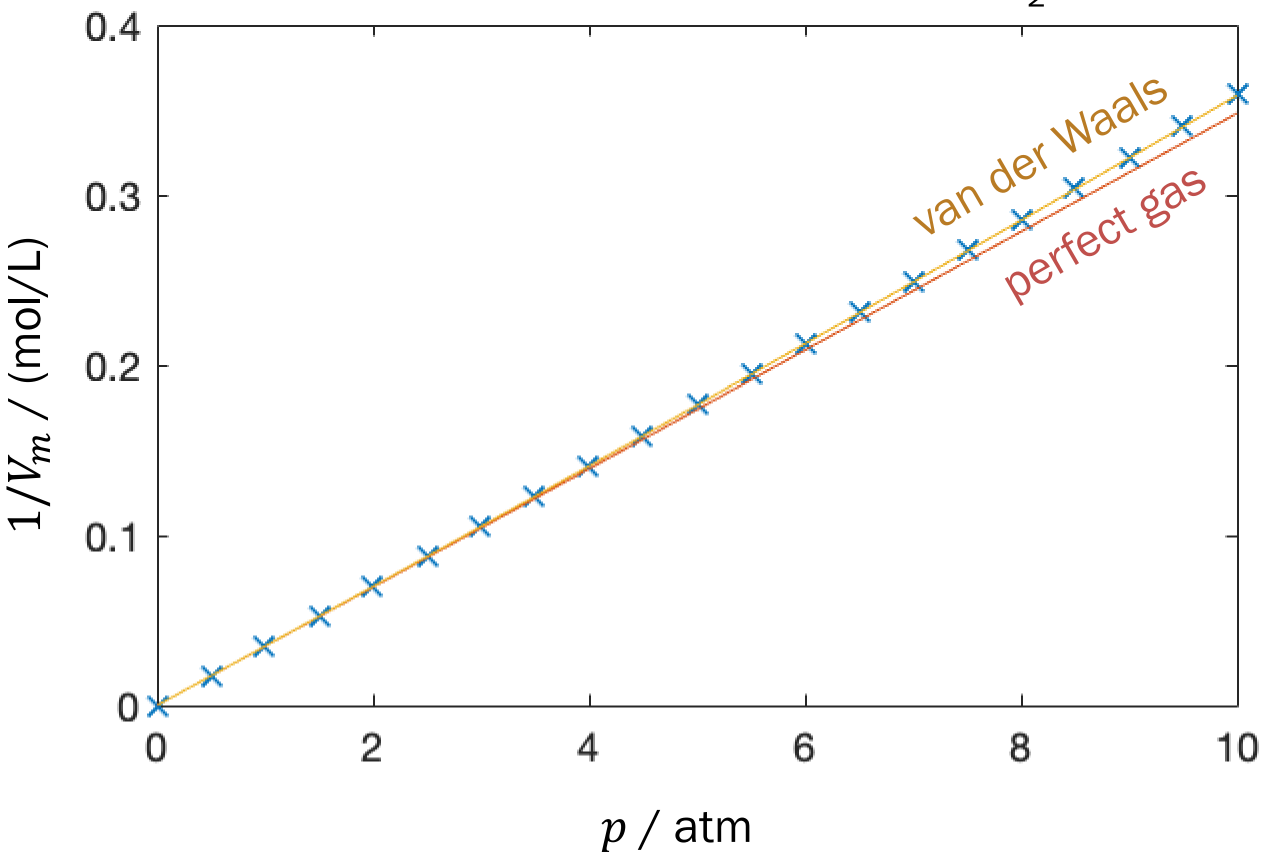
Validity of van der Waals

Data: CO_2 at 350 K



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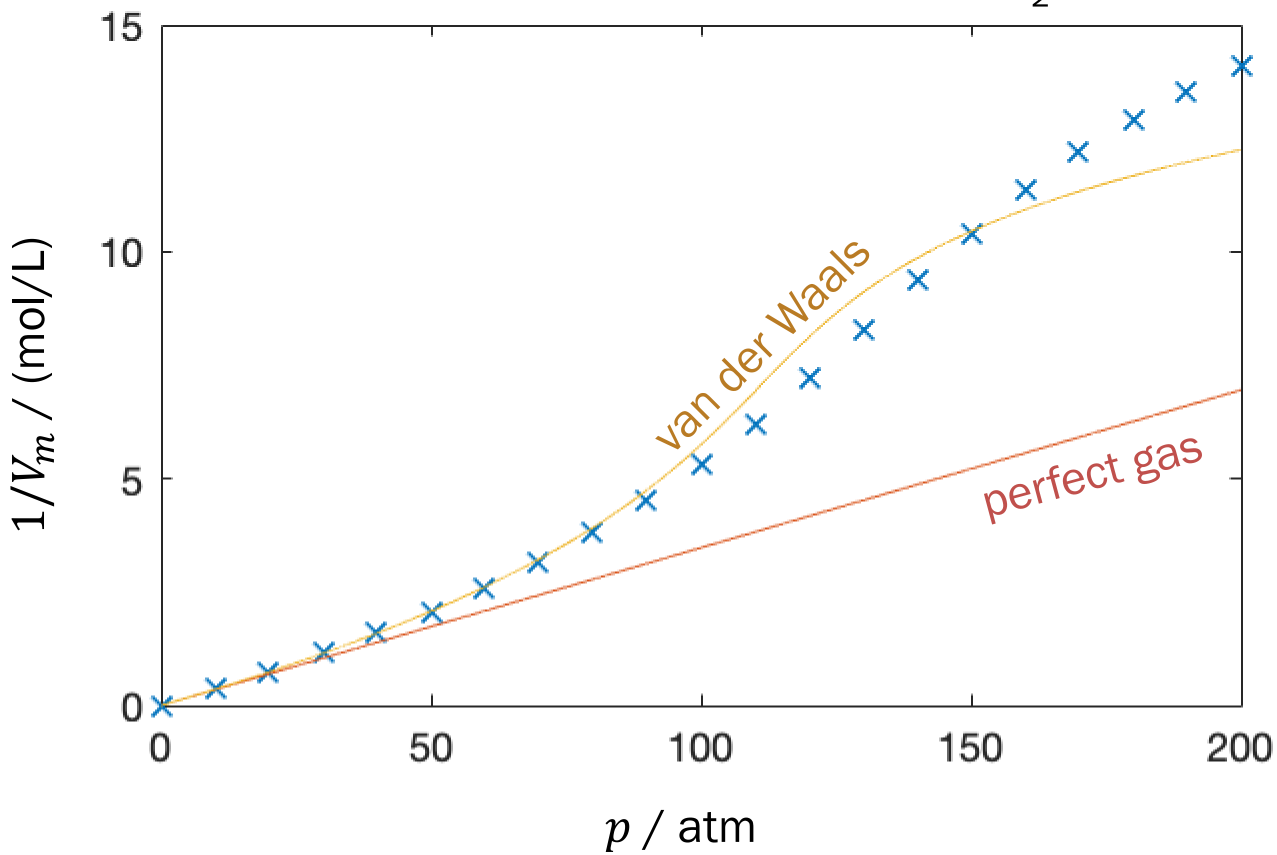
Is van der Waals Perfect?

Data: CO_2 at 350 K



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Other Real Gas Equations

- van der Waals equation (1873)

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

- Modification of the attraction term
  - Redlich-Kwong equation (1949)

$$p = \frac{RT}{V_m - B_{\mathrm{RK}}} - \frac{A_{\mathrm{RK}}}{V_m(V_m + B_{\mathrm{RK}})T^{1/2}}$$

  - Peng-Robinson equation (1976)

$$p = \frac{RT}{V_m - B_{\mathrm{PR}}} - \frac{A_{\mathrm{PR}}}{V_m(V_m + B_{\mathrm{PR}}) + B_{\mathrm{PR}}(V_m - B_{\mathrm{PR}})}$$

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Other Real Gas Equations

- van der Waals equation (1873)

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

- Modification of the repulsion term
  - Guggenheim equation (1965)

$$p = \frac{RT}{V_m(1-y)^4} - \frac{A_{\mathrm{G}}}{V_m^2}, \qquad y = \frac{B_{\mathrm{G}}}{V_m}$$

  - Carnahan-Starling-van der Waals equation (1972)

$$p = \frac{RT(1+y+y^2-y^3)}{V_m(1-y)^3} - \frac{A_{\mathrm{CSW}}}{V_m^2}, \qquad y = \frac{B_{\mathrm{CSW}}}{V_m}$$

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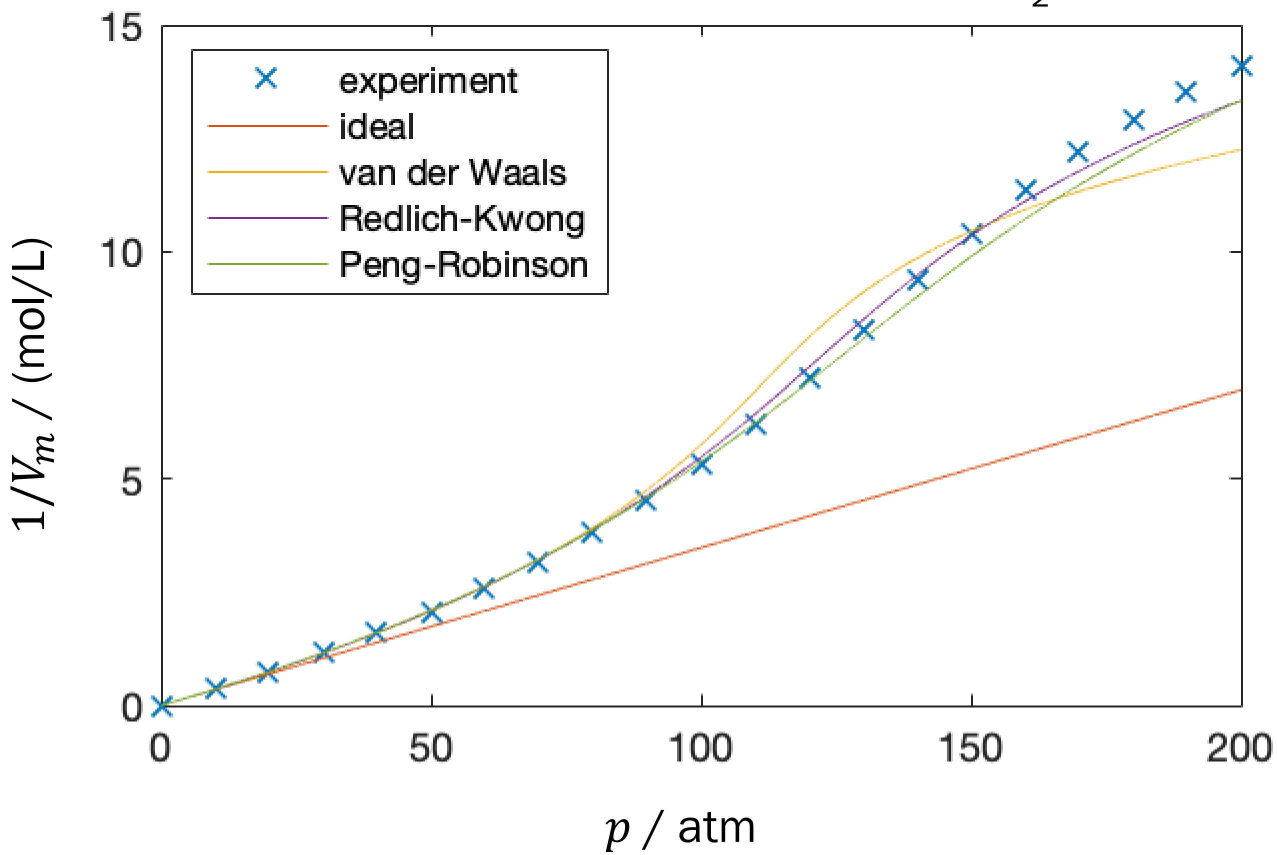
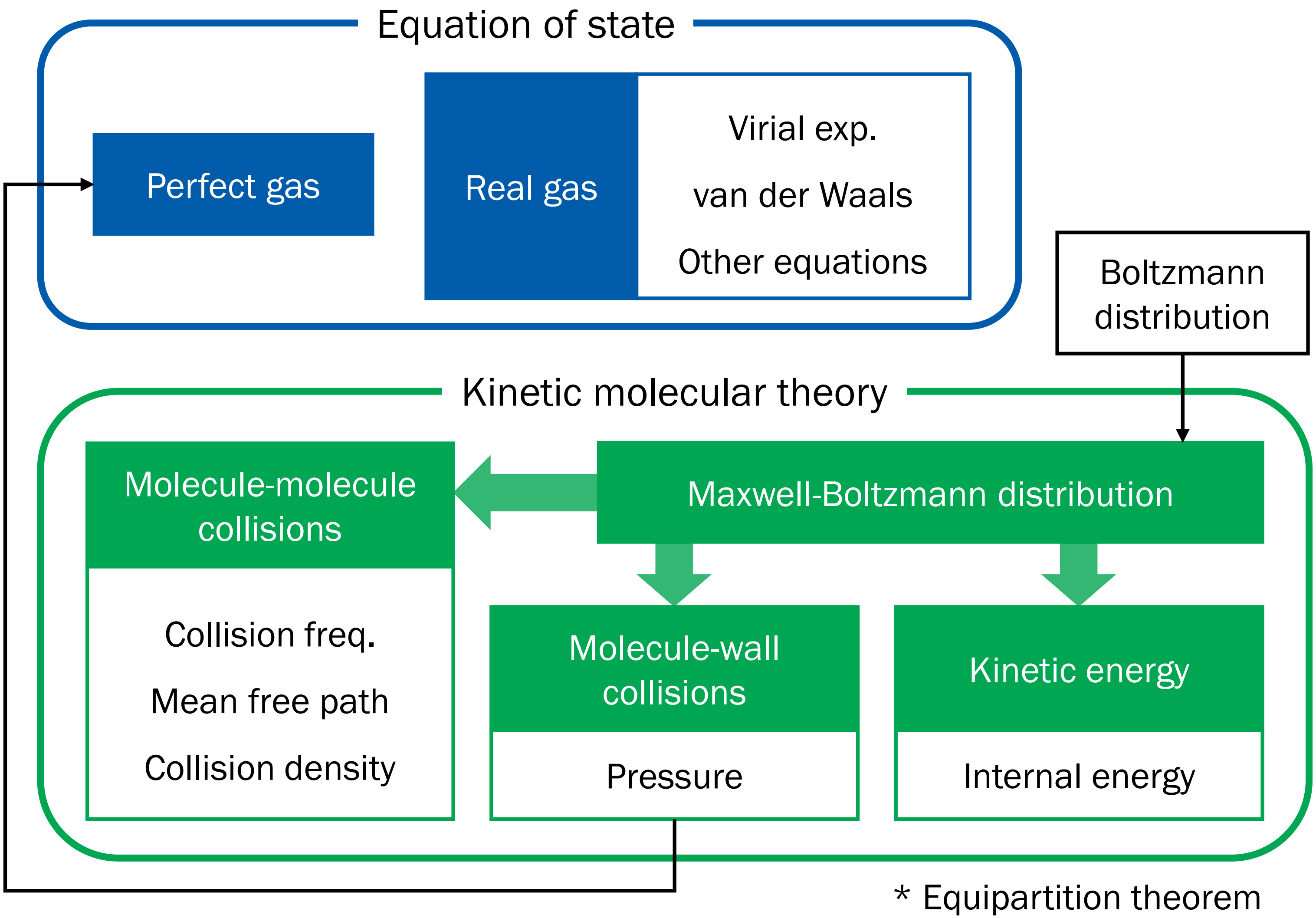
Better Descriptions

Data: CO_2 at 350 K



Summary


Equation of state
Perfect gas
Real gas
Virial exp.
van der Waals
Other equations
Boltzmann distribution
Kinetic molecular theory
Molecule-molecule collisions
Maxwell-Boltzmann distribution
Collision freq.
Mean free path
Collision density
Molecule-wall collisions
Pressure
Kinetic energy
Internal energy
* Equipartition theorem

3. The First Law

Chapter Overview

- First law of thermodynamics
  - Work and heat
  - Differential form of internal energy
- Enthalpy
  - Definition and properties of enthalpy
  - Differential form
  - Adiabatic change
- Thermochemistry

Internal Energy, U

"The total energy of a system"

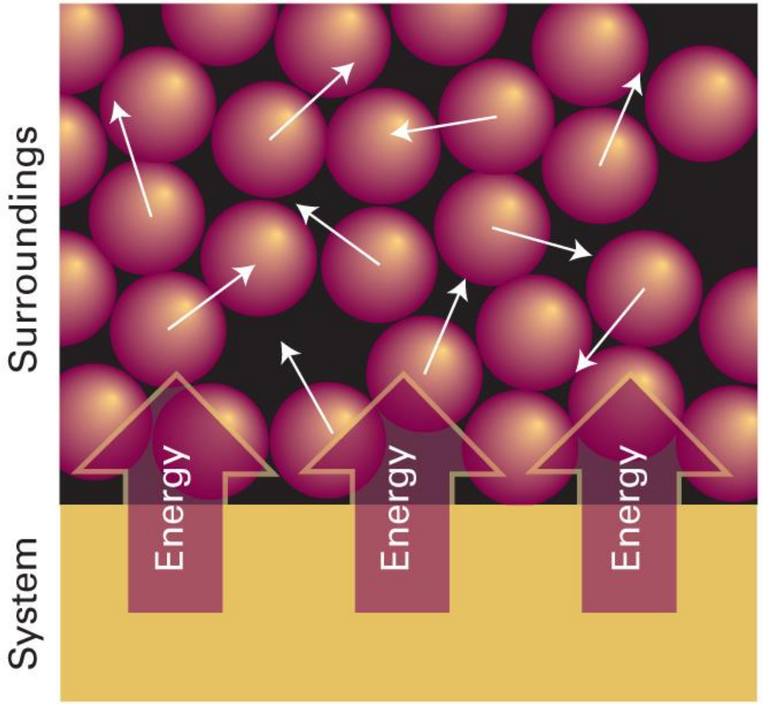
- Perfect gas (monatomic): $U = (3/2)nRT$
- Molar internal energy

$$U_m = U/n$$

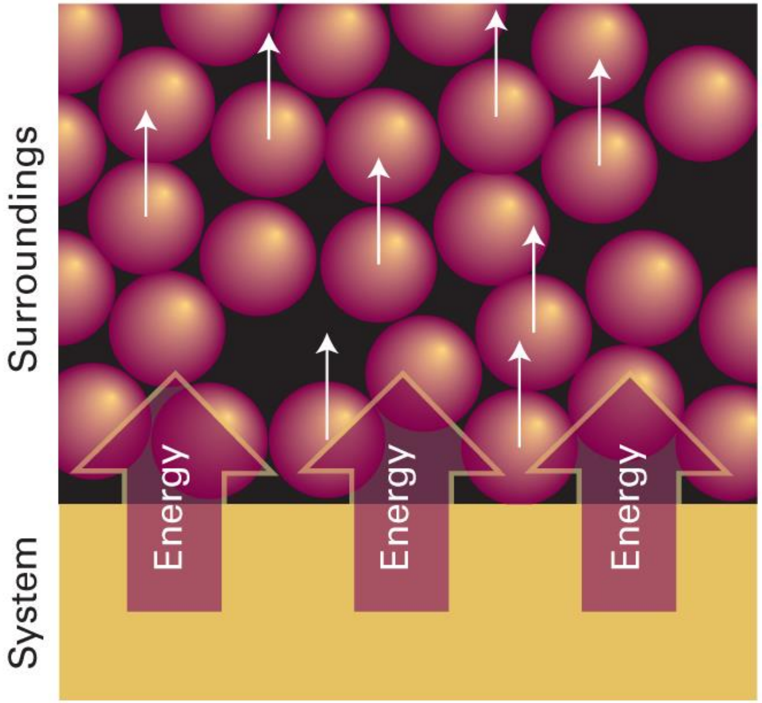
For a perfect gas, $U_m = (3/2)RT$

How can we change the internal energy?

Two Possible Ways



“Heat”



“Work”

(Image: Atkins *et al*., ***Atkins’ Physical Chemistry***, 11th ed., Figures 2A.3 and 2A.4)

First Law of Thermodynamics

"The internal energy of an isolated system is constant."

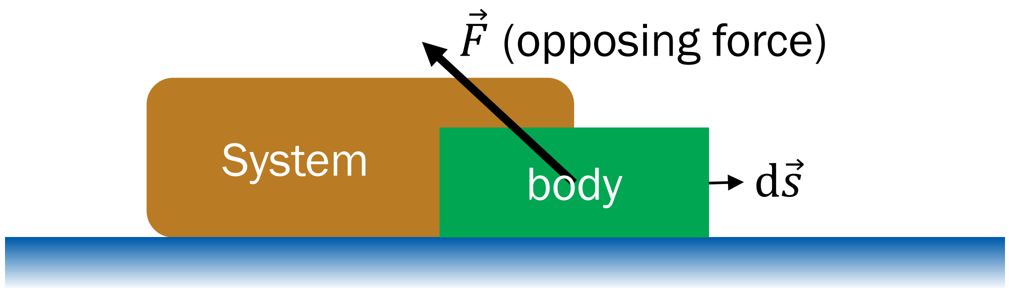
- Heat and work are equivalent ways of changing the internal energy of a system

$$\Delta U = q + w$$

$$\mathrm{d}U = \delta q + \delta w$$

small change of U small q small w

Work



- Work done on the body

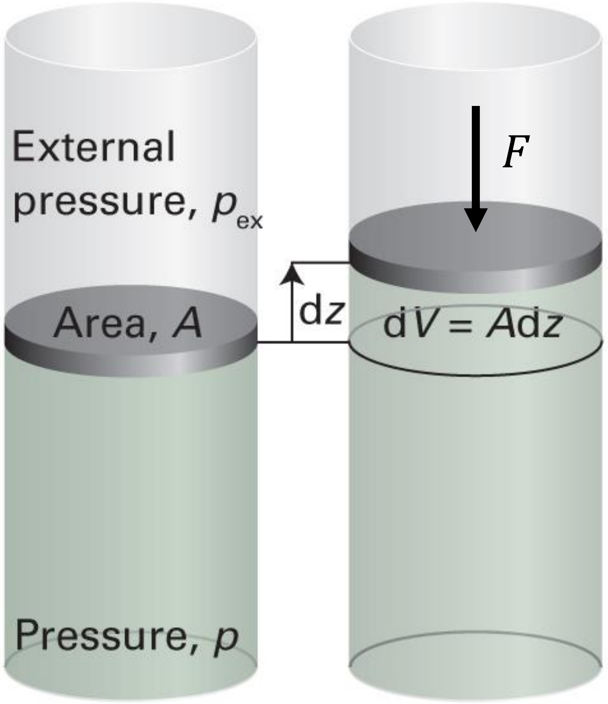
$$\delta w_{\mathrm{body}} = -\vec{F} \cdot \mathrm{d}\vec{s}$$

- Work done on the system

$$\delta w = \vec{F} \cdot \mathrm{d}\vec{s}$$

where $\vec{F} \cdot \mathrm{d}\vec{s} = F_x \mathrm{d}x + F_y \mathrm{d}y + F_z \mathrm{d}z$

Expansion Work



$$\delta w = \vec{F} \cdot \mathrm{d}\vec{s} = F_x \mathrm{d}x + F_y \mathrm{d}y + F_z \mathrm{d}z$$

As $\mathrm{d}x = \mathrm{d}y = 0$,

$$\delta w = -|F|\mathrm{d}z$$

Since $p_{\mathrm{ex}} = |F|/A$,

$$\delta w = -p_{\mathrm{ex}}A\mathrm{d}z = -p_{\mathrm{ex}}\mathrm{d}V$$

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 2A.5)

Other Types of Work

$\delta w = $ (intensive factor) × (extensive factor)

Type of work	dw	Comments	Units†
Expansion	$-p_{ex}\mathrm{d}V$	p_{ex} is the external pressure	Pa
		dV is the change in volume	m^3
Surface expansion	$\gamma \mathrm{d}\sigma$	γ is the surface tension	$N\ m^{-1}$
		dσ is the change in area	m^2
Extension	$f\mathrm{d}l$	f is the tension	N
		dl is the change in length	m
Electrical	$\phi \mathrm{d}Q$	ϕ is the electric potential	V
		dQ is the change in charge	C
	$Q\mathrm{d}\phi$	dϕ is the potential difference	V
		Q is the charge transferred	C

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11^{th} ed., Table 2A.1)

Topic 2A

Calculation of Expansion Work

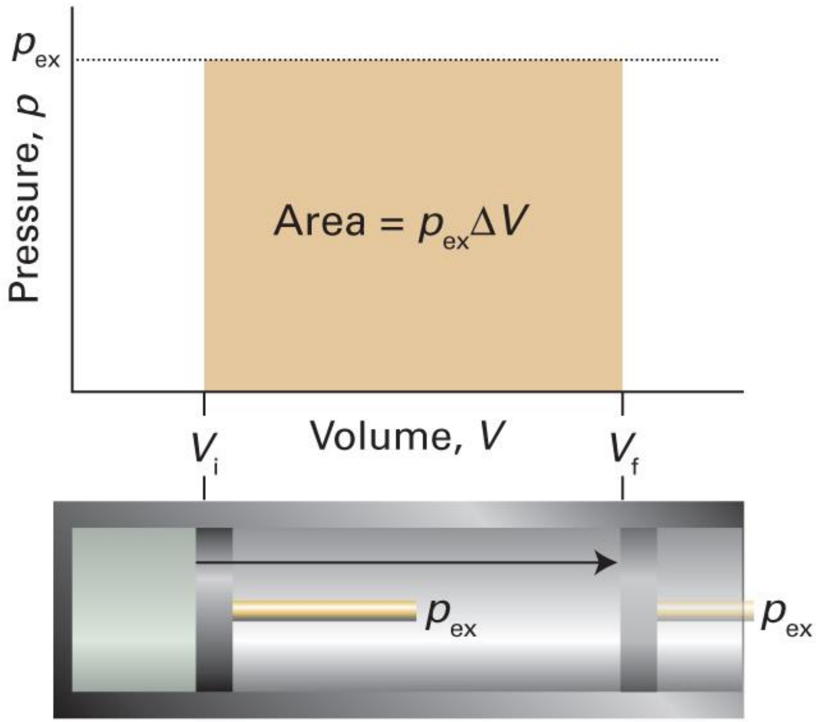
$$\delta w = -p_{\mathrm{ex}}\mathrm{d}V$$

- Total work for a change from state i to state f

$$w = \int_{\text{state } i}^{\text{state } f} \delta w = \int_{\text{state } i}^{\text{state } f} (-p_{\mathrm{ex}}\mathrm{d}V) = -\int_{V_i}^{V_f} p_{\mathrm{ex}}\,\mathrm{d}V$$

1. Expansion against constant external pressure
2. Reversible expansion

Constant External Pressure



$$w = -\int_{V_i}^{V_f} p_{ex}\,\mathrm{d}V$$

$$w = -p_{ex}\int_{V_i}^{V_f} \mathrm{d}V$$

$$w = -p_{ex}(V_f - V_i)$$

For $\Delta V = V_f - V_i$,

$$w = -p_{ex}\Delta V$$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 2A.6)

Free Expansion

$$w = -p_{\mathrm{ex}}\Delta V$$

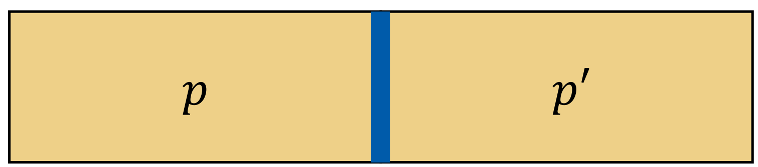
If $p_{\mathrm{ex}} = 0$, $w = 0$

(regardless of the value of ΔV)

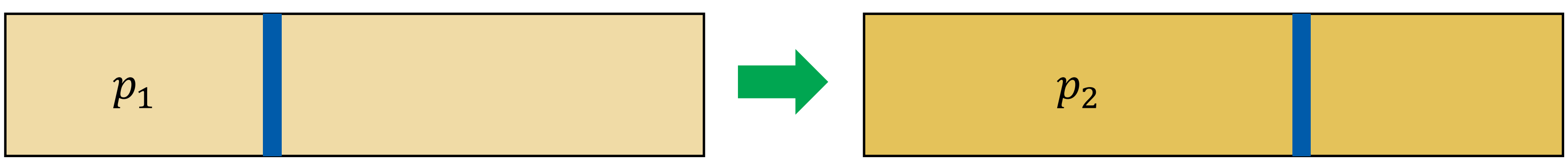
e.g. a gas expanding into a vacuum

Reversible Expansion

- Reversible change: a change that can be reversed by an infinitesimal modification of a variable
- The system is at equilibrium at any time during the change



$p = p'$ always



Reversible Expansion Work

$$w = -\int_{V_i}^{V_f} p_{\mathrm{ex}}\,\mathrm{d}V$$

For the reversible expansion, $p = p_{\mathrm{ex}}$ always.

$$w = -\int_{V_i}^{V_f} p\,\mathrm{d}V$$

Case of Perfect Gas

$$w = -\int_{V_i}^{V_f} p\,\mathrm{d}V$$

For the reversible expansion of a perfect gas,

$$pV = nRT$$

$$p = nRT/V$$

Hence,

$$w = -\int_{V_i}^{V_f} p\,\mathrm{d}V = -\int_{V_i}^{V_f} \frac{nRT}{V}\mathrm{d}V = -nR\int_{V_i}^{V_f} \frac{T}{V}\mathrm{d}V$$

Topic 2A

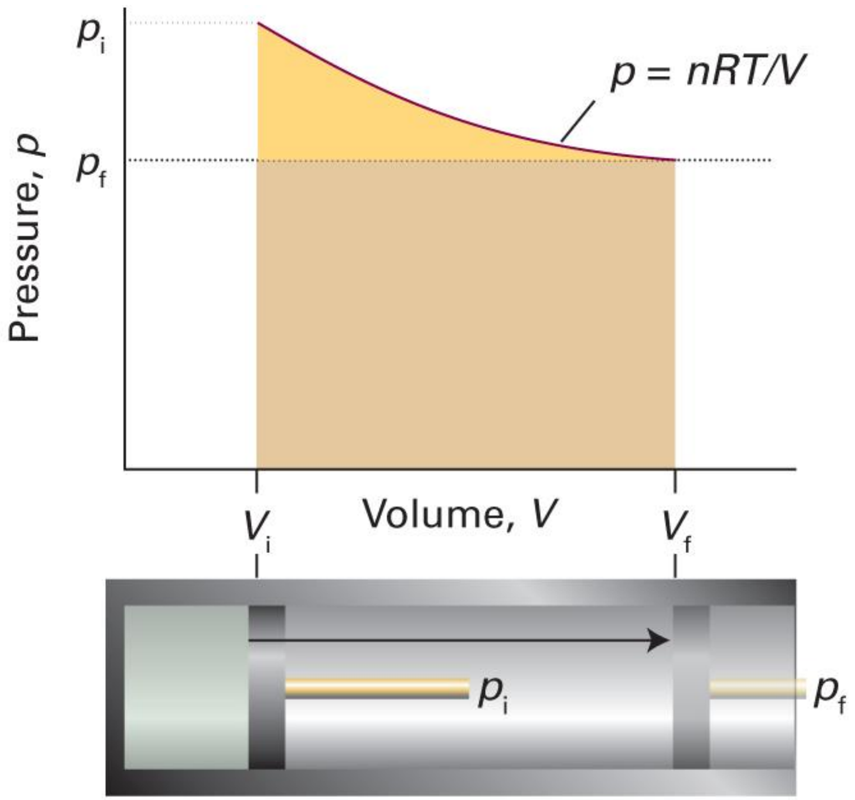
Perfect Gas, Constant T

For an isothermal reversible expansion,

$$w = -nR\int_{V_i}^{V_f} \frac{T}{V}\,\mathrm{d}V = -nRT\int_{V_i}^{V_f} \frac{1}{V}\,\mathrm{d}V$$

$$w = -nRT\left(\ln V_f - \ln V_i\right) = -nRT\ln\frac{V_f}{V_i}$$

Perfect Gas, Constant T



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 2A.7)

Heat

$$q \propto \Delta T$$

For infinitesimal $\mathrm{d}T$,

$$\delta q \propto \mathrm{d}T$$

$$\delta q = C\mathrm{d}T$$

(C: heat capacity)

Calculation of Heat

$$\delta q = C\mathrm{d}T$$

- Total heat for a change from state i to state f

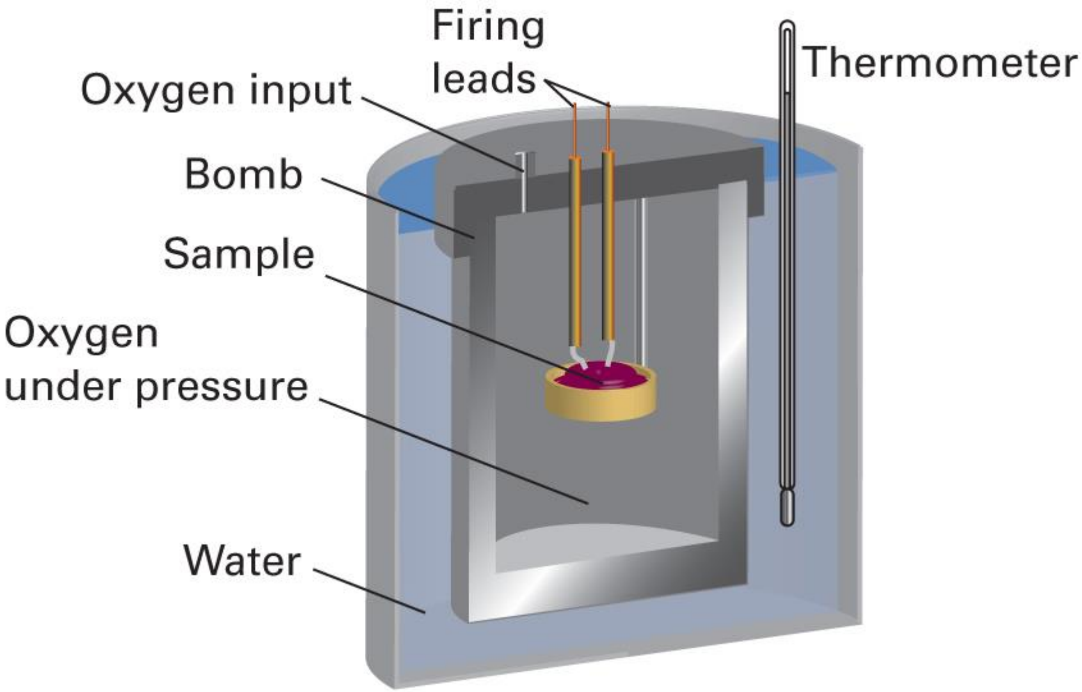
$$q = \int_{\text{state } i}^{\text{state } f} \delta q = \int_{\text{state } i}^{\text{state } f} (C\mathrm{d}T) = \int_{T_i}^{T_f} C\,\mathrm{d}T$$

Heat Capacity

- Molar heat capacity $C_m = C/n$
- Specific heat capacity $C_s = C/m$

- Heat capacity depends on the constraint
  - C_V: heat capacity at constant volume ($C_{V,m}$, $C_{V,s}$)
  - C_p: heat capacity at constant pressure ($C_{p,m}$, $C_{p,s}$)

Calorimetry: Heat Measurement



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 2A.8)

First Law Revisited

$$\mathrm{d}U = \delta q + \delta w$$

$$\mathrm{d}U = \delta q + \delta w_{\mathrm{exp}} + \delta w_{\mathrm{add}}$$

$$\mathrm{d}U = C\mathrm{d}T - p_{\mathrm{ex}}\mathrm{d}V + \delta w_{\mathrm{add}}$$

For $\delta w_{\mathrm{add}} = 0$,

$$\mathrm{d}U = C\mathrm{d}T - p_{\mathrm{ex}}\mathrm{d}V$$

At Constant V

$$\mathrm{d}U = C\mathrm{d}T - p_{\mathrm{ex}}\mathrm{d}V$$

At constant volume, $\mathrm{d}V = 0$ and $C = C_V$

$$\mathrm{d}U = C_V\mathrm{d}T$$

The change in internal energy is

$$\Delta U = \int_{\text{state } i}^{\text{state } f} \mathrm{d}U = \int_{\text{state } i}^{\text{state } f} (C_V\mathrm{d}T) = \int_{T_i}^{T_f} C_V\,\mathrm{d}T$$

At Constant V

$$\Delta U = \int_{T_i}^{T_f} C_V \, \mathrm{d}T$$

If C_V is constant,

$$\Delta U = \int_{T_i}^{T_f} C_V \, \mathrm{d}T = C_V \int_{T_i}^{T_f} \mathrm{d}T = C_V\left(T_f - T_i\right) = C_V \Delta T$$

Since $\Delta U = q + w = q$ at constant volume,

$$q = C_V \Delta T$$

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(Monatomic) Perfect Gas

$$U = \frac{3}{2} nRT$$

$$\mathrm{d}U = \frac{3}{2} nR\mathrm{d}T$$

$$\mathrm{d}U = C\mathrm{d}T - p_{\mathrm{ex}}\mathrm{d}V$$

$$\frac{3}{2} nR\mathrm{d}T = C\mathrm{d}T - p_{\mathrm{ex}}\mathrm{d}V$$

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Perfect Gas at Constant V

$$\frac{3}{2}nR\mathrm{d}T = C\mathrm{d}T - p_{\mathrm{ex}}\mathrm{d}V$$

At constant volume, $\mathrm{d}V = 0$ and $C = C_V$

$$\frac{3}{2}nR\mathrm{d}T = C_V\mathrm{d}T$$

Hence,

$$C_V = \frac{3}{2}nR$$